# Self-Preferencing and Welfare in Hybrid Platforms

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# Hybrid Platforms and Self-Preferencing

- E-commerce platforms combine 1P and 3P, e.g. Amazon, JD.com, Walmart, Target+, Best Buy, etc.
  - 1P listings: limited in variety.
  - 3P listings: expand variety, but crowd out 1P sales.
- Platforms may want to **self-preference (SP)**, i.e., lower the visibility of 3P listings, to protect 1P sales.

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- Platforms may want to **self-preference (SP)**, i.e., lower the visibility of 3P listings, to protect 1P sales.
- Self-preferencing is a global policy concern:

  EU DMA explicitly prohibits SP; Amazon etc. under investigation
  CN Anti-Monopoly Guidelines on Platforms target SP
  JP Smartphone Competition Act (2024) addresses platform SP
  US FTC antitrust case against Amazon Buy Box SP

## Model:

- A simple model of a monopolist platform that combines first and third party sales, featuring:
  - **Free entry of 3P sellers**
  - **Control of the visibility of listings on the platform**

## Research Questions:

- When does the platform profit from self-preferencing?
- Welfare consequences of banning SP?

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  - $\Rightarrow$ Higher double marginalization (welfare deteriorates)
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- A ban on SP has ambiguous welfare effects:
- The platform responds by raising commission fees:
  - $\Rightarrow$ Higher double marginalization (welfare deteriorates)
  - $\Rightarrow$ Seller entry may increase *or* decrease (ambiguous welfare effects)
- We identify conditions under which a ban **lowers / increases** both consumer surplus and welfare.

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- **Platform**:
  - Platform charges commission fee $\gamma \in (0, 1)$ on 3P sales.
  - Owns measure $h > 0$ of **1P listings** ($h$ is exogenous)
  - Each listing is a unique variant at constant mc $c$

# Set-up (continue)

- The platform uses an algorithm to recommend products to consumers. Then, sellers set a price $p$ to maximize profits.
- The recommendation technology is given by

$$M(b, h+s)$$

  - $M$ gives successful recommendations.
  - $M(\cdot)$ is twice differentiable, increasing in both arguments.
- Each listing obtains $\mu^s$ buyers:

$$\mu^s(b, h+s) \equiv M(b, h+s)/(h+s).$$

  - $\mu^s(b, s)$ is increasing in $b$ and decreasing in $h+s$.

# Discussions of $M(\cdot)$

- Modeling SP: lower the likelihood that a 3P seller enters the recommendation process from 1 to $\alpha < 1$: $M(b, h + \alpha s)$
- Focus on the **visibility channel**
  - Motivated by platforms where visibility is a prerequisite for sales and often outweighs price competition.
  - Examples: Amazon's Buy Box, Booking.com's Rankings.
- Assume no direct pricing channel
  - We abstract away from using fees ($\gamma$) purely to raise 3P prices and divert demand to 1P.

# Micro-foundation for $M(\cdot)$ (I)

- A space of $N$ potential variants. On the platform, $H$ products are 1P listings, $S$ are 3P listings; $H + S < N$.
- Each buyer has a consideration set $\Omega$ (size $\omega$)
- The probability that the algorithm can find at least one variant in $\Omega$ among the $H + S$ variants:

$$\mu^b(H+S) = \zeta\left(1 - (1 - (H+S)/N)^{\omega}\right),$$

  where $\zeta \in (0, 1]$: the algorithm efficiency.
- In the large market limit ($N \to \infty$, $H/N \to h, S/N \to s$),

$$\mu^b(h+s) = \zeta\left(1 - e^{-(h+s)\omega}\right), \quad M(\cdot) = b \cdot \mu^b$$
$$\mu^s(b, h+s) = b \cdot \zeta\left(1 - e^{-(h+s)\omega}\right)/(h+s).$$

# Micro-foundation for $M(\cdot)$ (II)

- There are $B$ buyers, $H+S$ listings. For any buyer-listing pair, matching Prob. $q$ is i.i.d. drawn from a distribution $F(q)$ on $[0,1]$.
- After $q$'s are realized, the platform recommends the listing with:

$$q_{max} = \max_i \{q_i\}.$$

- Ex-ante, the expected matching Prob. for a buyer:

$$\mathbb{E}[q_{max}] = 1 - \int_0^1 [F(q)]^{H+S} dq.$$

- Total successful recommendations:

$$M(B, H+S) = B \cdot \left(1 - \int_0^1 [F(q)]^{H+S} dq\right)$$

- Treat $B, H, S$ as integers, and so have to deal with the integer issue for binding constraints.

## Timing

1. The platform announces the commission rate $\gamma$ (ad valorem).
2. Observing $\gamma$ and $M(\cdot)$, 3P sellers simultaneously decide whether to enter the platform.
3. The platform uses algorithm $M$ to recommend products to buyers.
4. Sellers (1P and 3P) set prices $p$; matched buyers purchase $D(p)$.

## Solution concept:

- Subgame perfection

# No Self-Preferencing

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- Define $\hat{\gamma}$: the rate that maximizes the <u>per-match fee revenue</u>

$$\hat{\gamma} \equiv \arg\max_\gamma \pi_P(\gamma).$$

# The Platform's Problem

- The free-entry condition for 3P sellers

$$\mu^s(b, h+s) \cdot \pi_S(\gamma) = k$$

pins down the measure of entering sellers: $s = s_0(\gamma)$, satisfying

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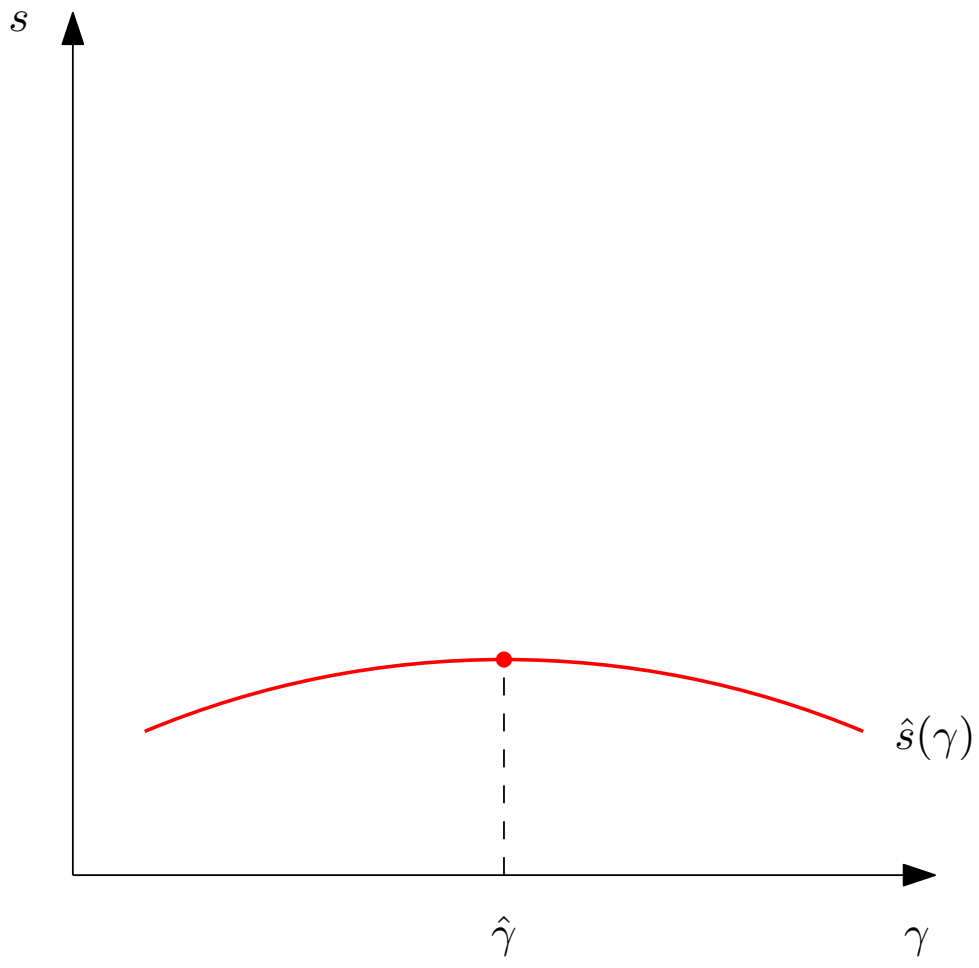
- The platform chooses commission rate $\gamma$ to maximize profits subject to free-entry condition:

$$\max_{\gamma \in [0,1]} \quad M\Big(b, h+s\Big)\Big(\frac{h}{h+s}\pi_M + \frac{s}{h+s}\pi_P(\gamma)\Big),$$
$$\text{s.t.} \quad s = s_0(\gamma).$$

- Consider the unconstrained problem first:

$$\max_{\gamma,s} \underbrace{M\Big(b, h+s\Big)}_{\text{Trade Volume}} \Big( \underbrace{\frac{h}{h+s}\pi_M + \frac{s}{h+s}\pi_P(\gamma)}_{\text{Business-stealing}} \Big).$$

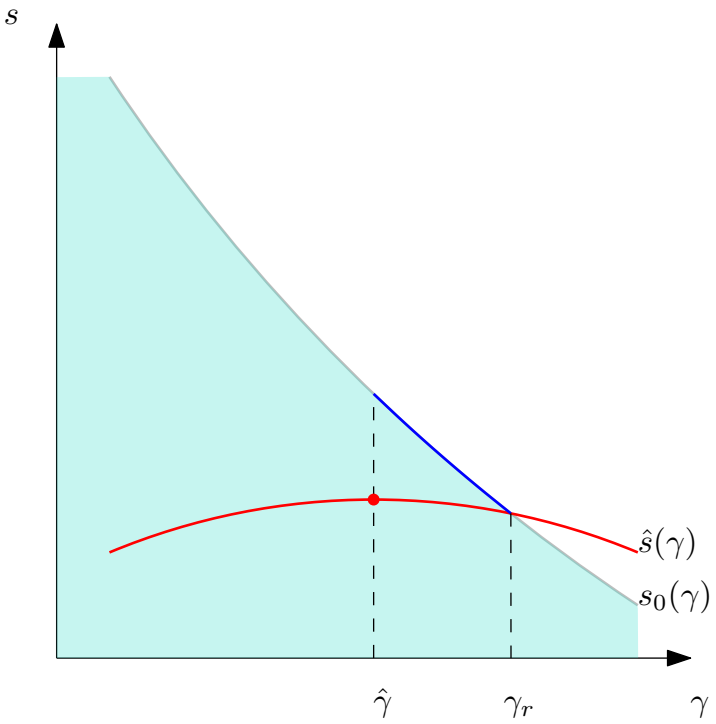
- Fix $\gamma$, and consider $s$ first:
  - Trade volume effect: $s \uparrow \rightarrow M(\cdot) \uparrow$.
  - Business stealing: $s \uparrow \rightarrow$ shift the sales mix toward less profitable 3P sellers (since $\pi_M > \pi_P(\gamma)$ for all $0 \leq \gamma \leq 1$).
- Assume the objective is single-peaked in $s$; solution: $\hat{s}(\gamma)$.
  - $\hat{s}(\gamma)$ is single-peaked in $\gamma$. When $\hat{s}(\gamma)$ is interior, it is increasing for $\gamma < \hat{\gamma}$ and decreasing for $\gamma > \hat{\gamma}$.




- The unconstrained optimum is $(\gamma^*, s^*) = (\hat{\gamma}, \hat{s}(\hat{\gamma}))$.

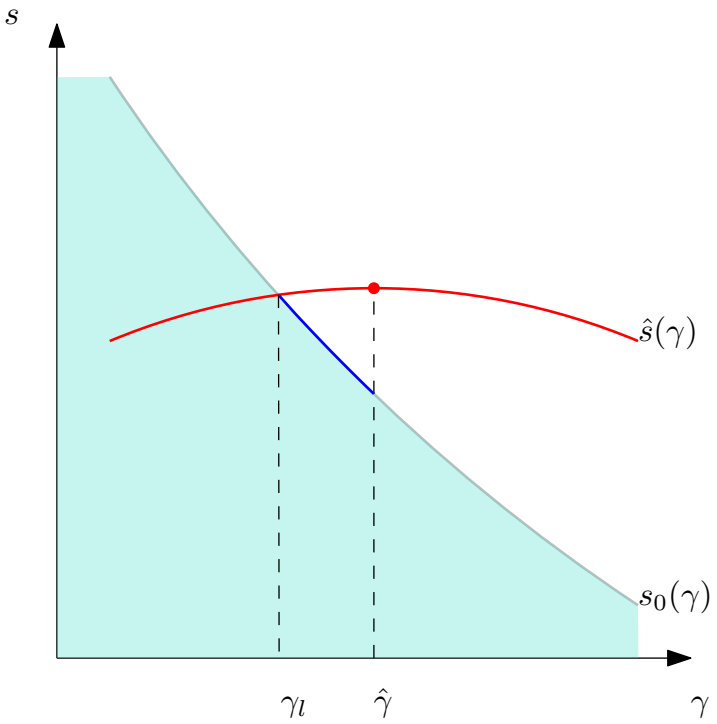
- Consider the constrained problem:

$$\max_{\gamma} \; M\Big(b, h+s\Big)\Big(\frac{h}{h+s}\pi_M + \frac{s}{h+s}\pi_P(\gamma)\Big), \text{ s.t. } s = s_0(\gamma).$$




$s_0(\hat{\gamma}) > \hat{s}(\hat{\gamma})$

**Excessive entry**




$s_0(\hat{\gamma}) < \hat{s}(\hat{\gamma})$

**Insufficient entry**

# Self-Preferencing

# Modeling SP

- SP: lower the likelihood that a 3P seller enters the recommendation process from 1 to $\alpha < 1$:

$$M(b, h + \alpha s)$$

- Choosing $\alpha$ is equivalent to choosing the number of displayed 3P sellers:

$$s_d = \alpha s.$$

This is clear by comparing:

$$\mu^s\left(b, h + \alpha s\right) \pi_{ss}(\gamma) \cdot \alpha = k$$
$$\mu^s\left(b, h + s_d\right) \pi_{ss}(\gamma) \cdot \frac{s_d}{s} = k.$$

# Modeling SP (cont.)

**Lemma**

*For any given $\gamma$, the platform can choose any $s_d \in [0, s_0(\gamma)]$. In particular, $s = s_0(\gamma)$ iff $s_d = s_0(\gamma)$; otherwise, $s > s_d$.*

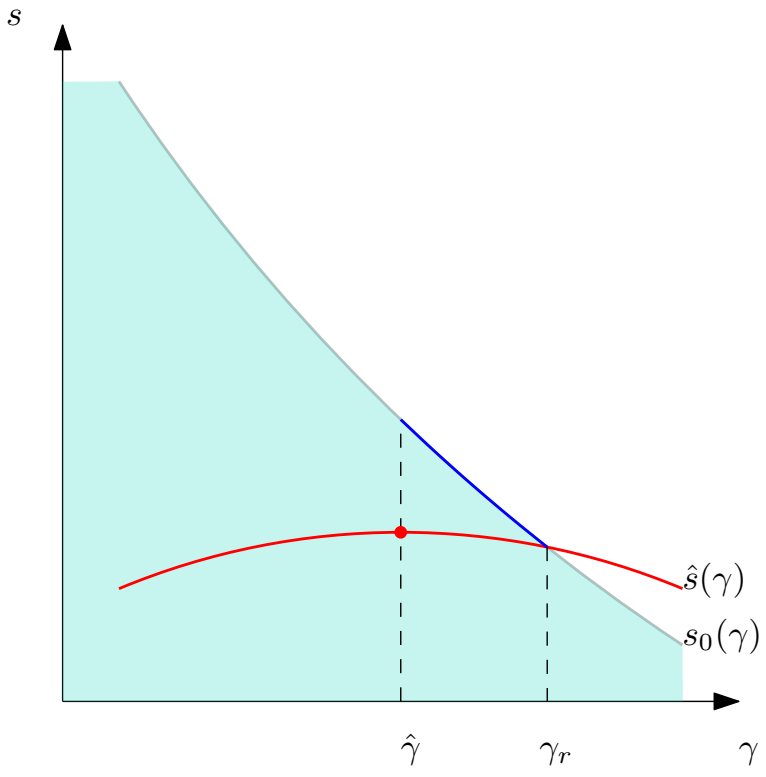
- SP gives the platform a tool to choose its ideal number of displayed sellers.
- In the rest analysis, we use $s_d$.

# Timing (updated)

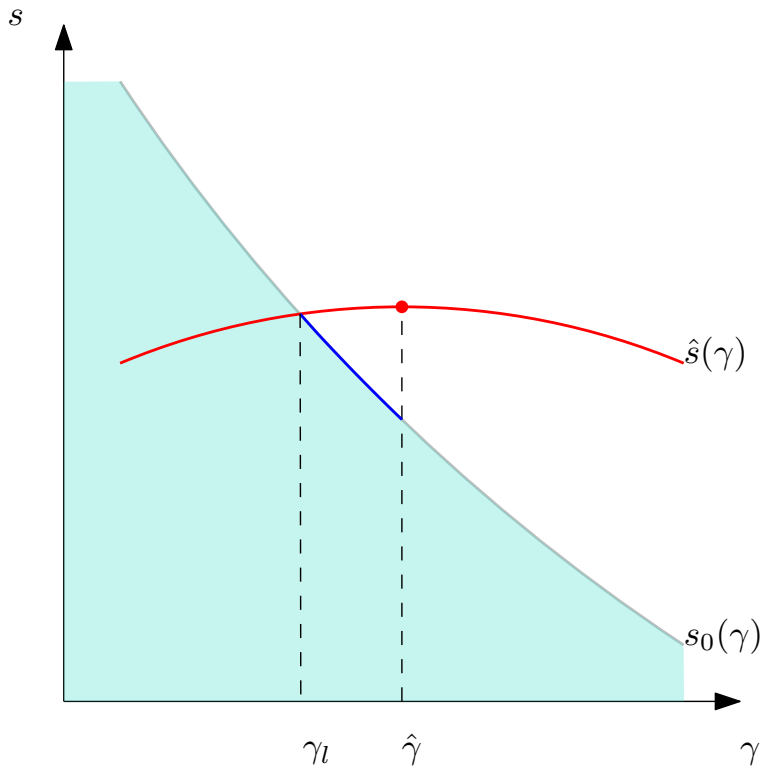
1. The platform announces the commission rate $\gamma$ and the number of displayed 3P sellers $s_d$.
2. Observing $(\gamma, s_d)$ and $M(\cdot)$, third-party sellers simultaneously decide whether to enter the platform.
3. The platform uses algorithm $M(b, h + s_d)$ to recommend products to buyers.
4. Sellers (1P and 3P) set prices $p$; matched buyers purchase the amount $D(p)$.

# The platform's problem (allowing for SP)

$$\max_{\gamma\in[0,1],s\in\mathbb{R}_+} M\Big(b, h+s\Big)\Big(\frac{h}{h+s}\pi_M + \frac{s}{h+s}\pi_P(\gamma)\Big), \text{ s.t. } s \leq s_0(\gamma).$$




$s_0(\hat{\gamma}) > \hat{s}(\hat{\gamma})$




$s_0(\hat{\gamma}) < \hat{s}(\hat{\gamma})$

## Proposition (Excessive Entry and SP)

*The platform's decision to self-preference depends on the level of entry at $\hat{\gamma}$.*

- **Excessive Entry ($s_0(\hat{\gamma}) > \hat{s}(\hat{\gamma})$):** *It is optimal for the platform to self-preference. The platform achieves unconstrained maximum profit by setting $\gamma_{sp} = \hat{\gamma}$ and $s_d = \hat{s}(\hat{\gamma})$.*
- **Insufficient Entry ($s_0(\hat{\gamma}) < \hat{s}(\hat{\gamma})$):** *It is optimal for the platform to* **not** *self-preference. The outcome is identical to the no self-preferencing benchmark.*

## Proposition (First-party Capacity triggers SP)

*If the platform's own capacity $h$ is sufficiently large ($h > \tilde{h}$), self-preferencing becomes profit-maximizing.*

- **Higher 1P capacity ($h \uparrow$)**: as 1P capacity increases, both seller entry $s_0(\hat{\gamma}) \downarrow$ and platform optimum $\hat{s}(\hat{\gamma}) \downarrow$.
- But the platform wants sellers to exit faster than they naturally do since 1P has a higher profit margin:

$$\underbrace{\frac{\partial s_0(\hat{\gamma})}{\partial h} = -1}_{\text{Seller Natural Exit}} \quad > \quad \underbrace{\frac{\partial \hat{s}(\hat{\gamma})}{\partial h}}_{\text{Platform Optimal Exit}}$$

## Corollary (Low Demand and Entry Cost Trigger SP)

*If the market demand ($b$) is sufficiently low, or if the seller entry cost ($k$) is sufficiently low, the platform engages in self-preferencing.*

- **Low Demand ($b \downarrow$):** As the market shrinks, the platform's desire for variety ($\hat{s}$) drops faster than the sellers' willingness to enter ($s_0$).
- **Low Entry Cost ($k \downarrow$):** Low costs trigger a flood of entrants ($s_0 \uparrow$), but the platform's ideal number of sellers ($\hat{s}$) is unchanged, creating a massive "excessive entry" gap.

# Welfare Analysis

# Social Planner's Choice

A social planner maximizes welfare (ignore the free entry condition)

$$\max_{\gamma,s} \quad \underbrace{M(\cdot)\left\{\frac{h}{h+s}\Big(\pi_M + cs_M\Big) + \frac{s}{h+s}\Big(\pi_P(\gamma) + cs_S(\gamma)\Big)\right\}}_{\equiv W(\gamma,s)}$$

- Fee distortion: $\hat{\gamma} > \gamma_w$ — platform's commission is too high.
  - $\hookrightarrow$ Raises 3P prices $\Rightarrow$ double marginalization $\Rightarrow$ lower consumer surplus.
- Quantity distortion: $\hat{s} \neq \hat{s}_w$ — platform's curation ignores consumer surplus.
  - $\hookrightarrow$ Platform trades off $\pi_M$ vs. $\pi_P(\gamma)$; planner trades off $w_M$ vs. $w_S(\gamma)$.

# Quantity Distortion

- Platform's choice of $\hat{s}$ driven by $\frac{\pi_P(\gamma)}{\pi_M}$
- Planner's choice of $\hat{s}_w$ driven by $\frac{cs_S(\gamma)+\pi_P(\gamma)}{cs_M+\pi_M}$

**Lemma**

*Fixing $\gamma$, the platform lists* **too few** *3P sellers, $\hat{s}(\gamma) < \hat{s}_w(\gamma)$, iff*

$$\frac{cs_M}{cs_S(\gamma)} < \frac{\pi_M}{\pi_P(\gamma)}.$$

*Conversely, the platform lists* **too many** *3P sellers when the inequality is reversed.*

# Welfare Impact of Banning SP

- Focus on excessive entry scenario $s_0(\hat{\gamma}) > \hat{s}(\hat{\gamma})$.
- Welfare **with SP**: $W_{sp} \equiv W(\hat{\gamma}, \hat{s}(\hat{\gamma}))$.
- Welfare after **banning SP**: $W_b \equiv W(\gamma_{nsp},\ s_0(\gamma_{nsp}))$.
- Welfare change of banning SP is $W_b - W_{sp}$:

$$\underbrace{W(\gamma_{nsp},\ s_0(\gamma_{nsp})) - W(\hat{\gamma},\ s_0(\gamma_{nsp}))}_{\text{Fee Effect}(<0)}$$
$$+ \underbrace{W(\hat{\gamma},\ s_0(\gamma_{nsp})) - W(\hat{\gamma},\ \hat{s}(\hat{\gamma}))}_{\text{Quantity Effect (ambiguous sign)}}.$$

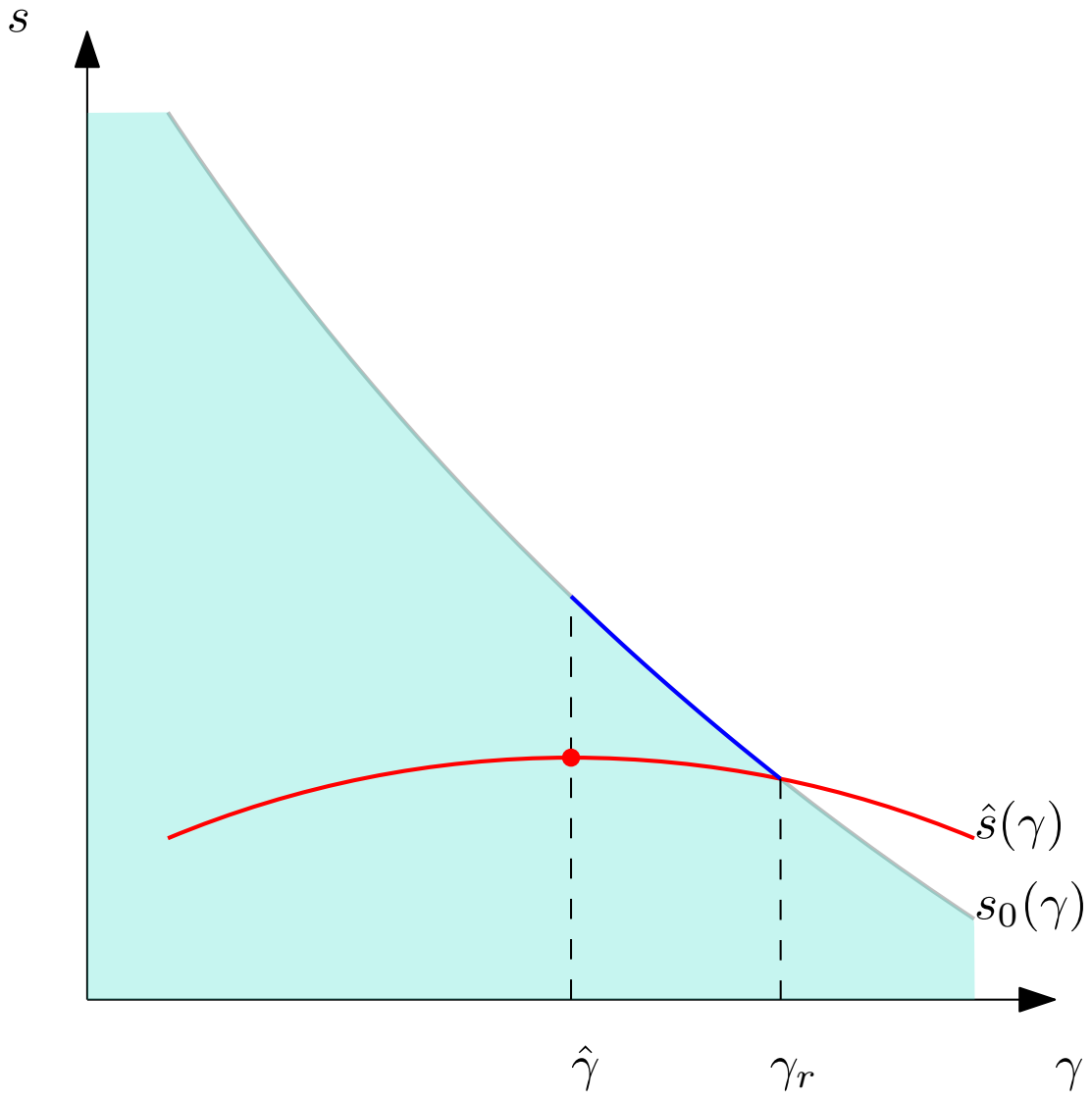
### Lemma

*There exist $h_0$ and $h_1$, satisfying $\tilde{h} < h_0 \leq h_1 < \overline{h}$, such that*

- *Banning SP leads to more entry ($\hat{s}(\hat{\gamma}) \leq s_0(\gamma_{nsp})$) if $h \leq h_0$;*
- *Banning SP leads to less entry ($\hat{s}(\hat{\gamma}) \geq s_0(\gamma_{nsp})$) if $h \geq h_1$.*

**Intuition:**

- **High** $h \Rightarrow$ **Over-deterrence:** The platform is very aggressive about protecting 1P sales, so the fee $\gamma_{nsp}$ is high enough to choke off entry.
- **Low** $h \Rightarrow$ **Insufficient deterrence:** The business stealing effect is smaller, so the platform uses a more moderate fee that fails to fully replicate the restrictive curation that SP would have allowed (leading to more entry).




$$s_0(\hat{\gamma}) > \hat{s}(\hat{\gamma})$$

- The Quantity Effect $W(\hat{\gamma},\ s_0(\gamma_{nsp})) - W(\hat{\gamma},\ \hat{s}(\hat{\gamma}))$ is negative if either:
  1. Platform lists too many sellers, a ban increases entry further
     $h < h_0$ and $\frac{cs_S(\hat{\gamma})}{cs_M} < \frac{\pi_P(\hat{\gamma})}{\pi_M} \Rightarrow s_0(\gamma_{nsp}) > \hat{s}(\hat{\gamma}) > s_w(\hat{\gamma})$
  2. Platform lists too few sellers, a ban reduces entry further
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- These conditions suffice for a ban to reduce both **welfare and consumer surplus**.
- **When is a ban beneficial?** If a ban moves variety closer to the social optimum (e.g., $s_w > s_0 > \hat{s}$) and this positive quantity effect is large enough to offset the fee distortion, then a ban improves welfare.

# Consumer Welfare Perspective

- **Consumer Surplus**:

$$CS(\gamma, s) = M(b, h+s)\left[\frac{h}{h+s} cs_M + \frac{s}{h+s} cs_S(\gamma)\right]$$

- The condition for the platform to list **too few** sellers ($\hat{s} < \hat{s}_c$) is identical to the social welfare case:

$$\frac{cs_M}{cs_S(\gamma)} < \frac{\pi_M}{\pi_P(\gamma)}$$

- The conditions are identical because both perspectives compare platform profit incentives ($\pi_M, \pi_P$) with user/social surplus ($cs_M, cs_S$).
- **Bottom Line**: Whenever a ban of SP reduces total welfare, it also reduces consumer welfare.

# Other Fee Structures

# Per-unit Fee

- Replace ad valorem fee with per-unit fee $\tau \geq 0$.
- Seller's Profit: $\pi_S(\tau) = \max_p [p - \tau - c] D(p)$
- Platform Revenue per match: $\pi_P(\tau) = \tau D(p_s(\tau))$
- Optimal measure of sellers $\hat{s}(\tau)$ is still single-peaked in $\tau$ and maximized at $\hat{\tau} \equiv \arg\max \pi_P(\tau)$.
- $\pi_S(\cdot)$, $\pi_P(\cdot)$, $s_0(\cdot)$, and $\hat{s}(\cdot)$ retain all essential properties (monotonicity, single-peakedness).
- This does not change the fundamental mechanics of self-preferencing or welfare analysis.

# Substitution between Fees and SP

The platform can hit a target variety $s' < s_0(\gamma)$ using either instrument:

- **Curation (SP)**: Deter surplus entry $\mathbf{s'_E} > \mathbf{s'}$ by limiting visibility:

$$\mu^s(h + s')\pi_S \cdot \frac{s'}{s'_E} = k$$

- **Entry Pricing ($F$)**: Deter entry directly by raising entry costs:

$$\mu^s(h + s')\pi_S = k + F$$

With $F = k\left(\frac{s'_E}{s'} - 1\right)$, both methods are outcome-equivalent for final variety.

# Superiority of Participation Fees

- While participation fee and SP can implement the same quantity of 3P sellers, they are **NOT** equivalent.
- In terms of **profit maximization**: Participation fee dominates SP.
  - **SP is wasteful**: Undisplayed entrants incur cost $k$ but create no matches. Fee is a direct transfer from sellers to the platform.
- In terms of **social welfare**:

$$W_F = W_{sp} + \hat{s} \cdot F > W_{sp}$$

  - The fee improves welfare by converting the **social waste** (redundant entry costs) into a **private transfer** to the platform.

# Summary

- **Entry Management**: SP emerges under excessive entry, decoupling fees from variety control.
- **Ban Risks**: Banning SP forces high commissions and triggers over-deterrence.
- **Welfare Driver**: $h$ is the critical determinant; a ban helps only when $h$ is low.

# Placing Our Work & Future Direction

# Literature Landscape

- **Pre-Match Self-Preferencing**
  - Platform intervenes before consumer preferences are revealed, applying uniform biases (e.g., blanket search weight reductions).
  - **Literature:** Anderson and Bedre-Defolie (2024), Zennyo (2022), Kittaka and Sato (2022).

- **Post-Match Self-Preferencing**
  - Platform intervenes after observing realized consumer matches or specific seller profitability (e.g., targeted steering).
  - **Literature:** de Cornière and Taylor (2019), Hagiu et al. (2022), Choi et al. (2023).

# A Thought Experiment

**Pre-Match SP**

(Unobservable Matches/Our Model)

- Uniformly lowers 3P visibility.
- Leads to lower trade volume.
- Fee is set to $\hat{\gamma}$.

**Post-Match SP**

(Ex-Post Targeted Steering)

- Demotes 3P *only* if a consumer matches 1P & 3P.
- Preserves matches (pure traffic transfer to 1P).
- 3P sellers become mere residual claimants. Platform must drop fee $\gamma$ **even lower**.

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**The Realistic "Interim" State**

Welfare outcomes are a function of the platform's information advantage.